3.2 Mittag-Leffler's Theorem.

Let $G \subset \mathbb{C}_\infty$ be an open set of the Riemann sphere

Let $P \subset G$ be a closed subset with no limit point in G (pole set)

For each $p \in P$, let $f_p \in \mathbb{C}[z]$ be a polynomial with no constant term

Then there exists a meromorphic function f on G such that :

- f is holomorphic on $G - P$
- for each $p \in P$,
$$\begin{cases} f(z) - f_p\left(\frac{1}{z-p}\right) & \text{is holomorphic on a neighborhood of } p \quad \text{if } p \neq \infty \\ f(z) - f_\infty(z) & \text{is holomorphic on a neighborhood of } \infty \quad \text{if } \infty \in P \end{cases}$$

i.e. on any open subset of $\mathbb{C}_\infty$, every **Additive** Cousin **Problem** has a solution

Proof. • Without loss of generality, assume $G \subset \mathbb{C}$ is an open set of the plane

• Apply the Compact Exhaustion Lemma to get a sequence $\{K_n\}$ of compact subsets of G such that (i) $G = \bigcup_{n=1}^{\infty} K_n$ (exhaustion)

(ii) for each n, $K_n \subset \operatorname{int} K_{n+1}$ (strictly increasing)

and (iii) for each n, every component of $\mathbb{C}_\infty - K_n$ contains a component of $\mathbb{C}_\infty - G$ (no extra holes)

• Choose a subset $E \subset \mathbb{C}_\infty - G$ that meets each component of $\mathbb{C}_\infty - G$. (eg. $E = \mathbb{C}_\infty - G$)

• For each n, the set $P_n := P \cap K_n$ is finite (because P has no limit point in compact $K_n \subset G$)

Define the rational function

$$F_n(z) := \sum_{p \in P_n - P_{n-1}} f_p\left(\frac{1}{z-p}\right) \qquad (P_0 := \phi)$$

(finite sum)

It is holomorphic on G except for possible poles at each $p \in P_n - P_{n-1}$ with principal part $f_p\left(\frac{1}{z-p}\right)$

eg. $U = G \setminus (P_n - P_{n-1})$

• Consider F_n as holomorphic function on some open set $U \subset G$ with $K_{n-1} \subset U$

By (iii), the set $E \subset \mathbb{C}_\infty - G \subset \mathbb{C}_\infty - U$ meets each (component of $\mathbb{C}_\infty - K_{n-1}$)

Hence **1.7 Runge's Theorem**

gives a rational function $R_n(z)$ whose poles lie only in E

such that $|F_n(z) - R_n(z)| < \frac{1}{2^n}$ uniformly for all $z \in K_{n-1}$

- By (i) & (ii), for any compact subset $K \subset G$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, one has $K \subset K_{n-1}$

Then $\sum_{n \geq N} \sup_{z \in K} \left| F_n(z) - R_n(z) \right|$ is $\leq \sum_{n \geq N} \frac{1}{2^n}$ which is $< \infty$

so $\sum_{n \geq N} \left(F_n(z) - R_n(z) \right)$ converges normally on K

- Since $K \subset G$ is arbitrary compact, this shows that

$$f(z) := \sum_{n=1}^{\infty} \left(F_n(z) - R_n(z) \right)$$ converges locally normally on $G - P$

- Thus f is a holomorphic function on $G - P$ with possible poles at each $p \in P$ with principal part $f_p\left(\frac{1}{z-p}\right)$

so f is a solution to the **Additive** Cousin **Problem** for G

#